Analysis: Cepheid Distances and the Hubble Constant

Determining the Expansion Rate of the Universe from Astrophysical Data

Author: Eivan Kolchin-Rivas

December 3, 2025

Abstract

This report presents a determination of the Hubble constant H_0 based on real Cepheid and galaxy data. The analysis proceeds in three stages: (1) calibration of the Cepheid Period Luminosity (PL) relation using Milky Way Cepheids with parallax distances; (2) determination of the distance to the galaxy NGC4527 from its Cepheid variables; and (3) estimation of H_0 by combining this distance with recession velocities and distances for seven additional galaxies. Standard statistical methods are used together with parameter decorrelation and intrinsic dispersion terms when required by the χ^2 values. The final result is

$$H_0 = 67.5 \pm 5.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}.$$

1 Overview of the Analysis Chain

The analysis is designed to mimic the "distance ladder" used in observational cosmology, but the focus here is purely on the data analysis steps rather than on the underlying astrophysics. The chain is:

1. **Step 1: Calibration:** Use 10 Milky Way Cepheids with parallaxes and extinction estimates to calibrate a visual band Period Luminosity (PL) relation.
2. **Step 2: Distance:** Apply this calibrated PL relation to 22 Cepheids in NGC4527 to determine the galaxy's distance modulus and distance.
3. **Step 3: Hubble Constant:** Combine the NGC4527 distance with its recession velocity, and then with equivalent measurements for seven additional galaxies, to estimate H_0 and assess the consistency of the dataset.

Notation and Conventions

Throughout, I use:

- m: apparent magnitude,
- M: absolute magnitude,
- $\mu = m - M$: distance modulus,
- d_{pc}: distance in parsecs (pc),
- D: distance in megaparsecs (Mpc),
- P: pulsation period in days,
- A: extinction in magnitudes,
- H_0: Hubble constant ($\mathrm{km\,s^{-1}\,Mpc^{-1}}$),

- σ_X: uncertainty on quantity X,
- χ^2: chi squared; χ^2_ν: reduced chi squared (ν = degrees of freedom),
- $\mathrm{corr}(x, y)$: correlation coefficient between fit parameters x and y,
- x_0: reference value of $\log_{10} P$ used to reduce parameter correlations.

All logarithms are base 10.

2 Step 1: The Cepheid Period Luminosity Relation

2.1 Data and absolute magnitudes

The Milky Way calibration sample consists of 10 Cepheids. For each star, the following are provided:

- parallax p in mas with uncertainty σ_p;
- pulsation period P (days);
- apparent visual magnitude m;
- extinction A (magnitudes) with uncertainty σ_A.

Uncertainties in P and m are assumed to be negligible compared to those in parallax and extinction and are ignored.
The distance in parsecs is determined from the parallax via

$$d_{\mathrm{pc}} = \frac{1000}{p_{\mathrm{mas}}}, \tag{1}$$

with uncertainty

$$\sigma_d = \frac{1000\,\sigma_p}{p^2}. \tag{2}$$

The absolute magnitude is then

$$M = m - 5\log_{10} d_{\mathrm{pc}} + 5 - A. \tag{3}$$

This is the standard conversion between apparent and absolute magnitude, with the extinction term $-A$ correcting for dimming along the line of sight.
To propagate uncertainties, I treat d_{pc} and A as independent. From Eq. (3),

$$\frac{\partial M}{\partial d_{\mathrm{pc}}} = -\frac{5}{\ln 10}\,\frac{1}{d_{\mathrm{pc}}}, \qquad \frac{\partial M}{\partial A} = -1. \tag{4}$$

Combining these gives

$$\sigma_M^2 = \left(-\frac{5}{\ln 10}\,\frac{\sigma_d}{d_{\mathrm{pc}}}\right)^2 + \sigma_A^2 = \left(\frac{5}{\ln 10}\,\frac{\sigma_p}{p}\right)^2 + \sigma_A^2, \tag{5}$$

so the error on the absolute magnitude is dominated by the parallax and extinction uncertainties.
The independent variable for the PL relation is

$$x = \log_{10} P, \qquad y = M. \tag{6}$$

2.2 Unshifted PL fit and statistical critique

The basic PL relation is modelled as a straight line

$$M = \alpha \log_{10} P + \beta. \tag{7}$$

I perform a weighted least squares fit, minimizing

$$\chi^2(\alpha, \beta) = \sum_i \frac{\left[y_i - (\alpha x_i + \beta)\right]^2}{\sigma_{M,i}^2}, \tag{8}$$

with weights $w_i = 1/\sigma_{M,i}^2$.
For this unshifted formulation, I obtain:

$$\alpha = -2.401 \pm 0.232, \tag{9}$$
$$\beta = -1.608 \pm 0.202. \tag{10}$$

The reduced chi squared is

$$\chi^2_\nu \simeq 0.535 \quad (\nu = 8), \tag{11}$$

and the correlation coefficient between α and β is

$$\mathrm{corr}(\alpha, \beta) \simeq -0.970. \tag{12}$$

A reduced χ^2 significantly below 1 implies that, on average, the points lie closer to the best fit line than the quoted uncertainties would suggest. In this case, $\chi^2_\nu \approx 0.535$ corresponds to the errors being larger than strictly required by a factor of roughly $\sqrt{1/0.535} \approx 1.3$. This likely reflects conservative parallax and extinction uncertainties. I keep these conservative errors, since they simply make subsequent parameter uncertainties slightly larger and do not indicate any problem with the linear PL model itself.

2.3 Shifted PL fit and parameter decorrelation

To reduce the correlation between slope and intercept, I reparametrise the PL relation as

$$M = a\,(\log_{10} P - x_0) + b, \tag{13}$$

where the reference value x_0 is chosen as the weighted mean of $x_i = \log_{10} P_i$:

$$x_0 = \frac{\sum_i w_i x_i}{\sum_i w_i}. \tag{14}$$

Numerically,

$$x_0 = 0.845. \tag{15}$$

Refitting in terms of $x' = \log_{10} P - x_0$ yields

$$a = -2.401 \pm 0.232, \tag{16}$$
$$b = -3.636 \pm 0.049, \tag{17}$$

with

$$\chi^2_\nu \simeq 0.535, \qquad \mathrm{corr}(a, b) \simeq 0.000. \tag{18}$$

As expected, the slope is unchanged, the fit quality is identical, and the correlation between the new parameters is essentially zero. In what follows, I use this shifted parametrisation, because the lack of correlation between a and b simplifies error propagation to other quantities.
Figure 1 illustrates the calibrated PL relation and the best fitting line.

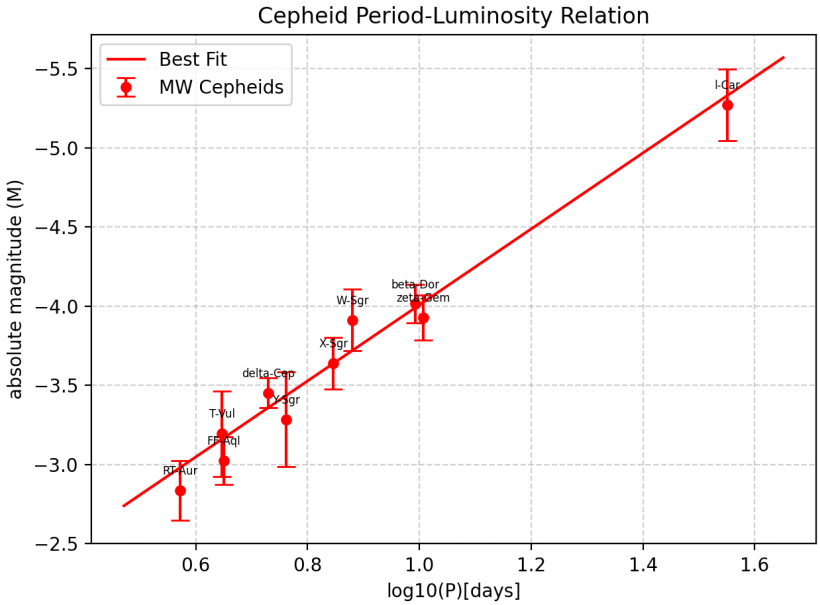


Figure 1: Cepheid Period Luminosity relation for the Milky Way calibration sample. Points show absolute magnitudes with 1σ error bars. The line shows the best fitting relation $M = a(\log_{10} P - x_0) + b$, with $a = -2.401$ and $b = -3.636$ at $x_0 = 0.845$.

3 Step 2: Distance to NGC4527

3.1 Cepheid distance moduli in NGC4527

The NGC4527 dataset contains 22 Cepheids, each with name, $\log_{10} P$ (days), and apparent magnitude m. The line of sight includes Milky Way foreground extinction

$$A_{\rm MW} = 0.0682\,\mathrm{mag}, \tag{19}$$

which I correct for uniformly. No star by star internal extinction corrections within NGC4527 are available, so any extra dimming within the galaxy is treated statistically.
For each Cepheid i in NGC4527, the predicted absolute magnitude from the PL relation is

$$M_i = a\,(\log_{10} P_i - x_0) + b, \tag{20}$$

with a, b and x_0 from Step 1. Because a and b are uncorrelated, the uncertainty on M_i is

$$\sigma^2_{M,i} = (\log_{10} P_i - x_0)^2\,\sigma_a^2 + \sigma_b^2. \tag{21}$$

The distance modulus of each Cepheid is then

$$\mu_i = m_i - A_{\rm MW} - M_i, \tag{22}$$

with $\sigma_{\mu,i} = \sigma_{M,i}$.

3.2 Intrinsic dispersion and galaxy distance

If all Cepheids in NGC4527 were perfect standard candles at exactly the same distance and fully corrected for extinction, the μ_i would scatter around a single value μ purely because of the propagated PL calibration errors. In practice, additional scatter is expected from internal extinction, geometric depth, and residual PL width.

I therefore fit a constant μ to the set $\{\mu_i\}$, but allow for an extra intrinsic dispersion term $\sigma_{\rm int}$ added in quadrature to each $\sigma_{\mu,i}$:

$$\sigma^2_{\mu,i,\rm tot} = \sigma^2_{\mu,i} + \sigma^2_{\rm int}. \tag{23}$$

The best fitting μ is found by minimizing the corresponding χ^2. I then adjust $\sigma_{\rm int}$ so that the reduced chi squared χ^2_ν is close to 1, as described in the course notes.
The resulting parameters are:

$$\mu = 30.768 \pm 0.072\,\text{mag}, \tag{24}$$

$$\sigma_{\rm int} \simeq 0.300\,\text{mag}, \tag{25}$$

$$\chi^2_\nu \simeq 0.960. \tag{26}$$

Thus the mean distance modulus is well constrained, while the intrinsic dispersion captures additional astrophysical scatter among the individual Cepheids.

3.3 Distance and its uncertainty

The distance modulus is related to the distance in parsecs by

$$\mu = 5\log_{10} d_{\rm pc} - 5 \quad \Rightarrow \quad d_{\rm pc} = 10^{(\mu+5)/5}. \tag{27}$$

With $\mu = 30.768$, this gives

$$d_{\rm pc} \simeq 1.424 \times 10^7 \text{ pc}, \tag{28}$$

or

$$D = \frac{d_{\rm pc}}{10^6} = 14.24 \text{ Mpc}. \tag{29}$$

To propagate the uncertainty in μ into an uncertainty in $d_{\rm pc}$, I use

$$\frac{{\rm d}d_{\rm pc}}{{\rm d}\mu} = \frac{\ln 10}{5}\, d_{\rm pc}, \tag{30}$$

so

$$\sigma_d = \frac{\ln 10}{5}\, d_{\rm pc}\, \sigma_\mu. \tag{31}$$

With $\sigma_\mu = 0.072\,\text{mag}$, this yields

$$D_{\rm NGC4527} = 14.24 \pm 0.48 \text{ Mpc}. \tag{32}$$

This value and its uncertainty are used in the H_0 determination.

4 Step 3: The Hubble Constant

4.1 Individual Hubble constant estimates

For NGC4527, the recession velocity is

$$v_{\rm rec} = 1152 \text{ km s}^{-1}, \tag{33}$$

with negligible instrumental uncertainty. The Hubble constant estimate from this galaxy alone is

$$H_{0,\rm NGC4527} = \frac{v_{\rm rec}}{D_{\rm NGC4527}} \simeq \frac{1152}{14.24} \simeq 80.9 \text{ km s}^{-1}\,\text{Mpc}^{-1}. \tag{34}$$

The corresponding error from the distance uncertainty is

$$\sigma_{H_0,\rm NGC4527} = \frac{v_{\rm rec}}{D^2}\,\sigma_D \simeq 2.7 \text{ km s}^{-1}\,\text{Mpc}^{-1}. \tag{35}$$

To improve constraints and assess consistency, I combine this with seven additional galaxies from `other_galaxies.dat`. For each galaxy i with recession velocity v_i, distance D_i, and distance error $\sigma_{D,i}$, I compute

$$H_{0,i} = \frac{v_i}{D_i}, \qquad \sigma_{H_0,i} = \frac{v_i}{D_i^2}\,\sigma_{D,i}. \tag{36}$$

This weighting scheme effectively scales as $w_i \propto D_i^2$ when the dominant additional scatter comes from peculiar velocities. Physically, if galaxies have a typical peculiar velocity $v_{\rm pec}$, the fractional contamination of the Hubble flow is $v_{\rm pec}/v_{\rm rec} \propto 1/D$. More distant galaxies therefore provide cleaner measurements of H_0, and giving them higher weight is well motivated.

4.2 Combining H_0 with intrinsic dispersion

If I simply compute a weighted mean of the eight $H_{0,i}$ using only the $\sigma_{H_0,i}$, I obtain a very small formal error but an unacceptably large reduced chi squared ($\chi^2_\nu \gg 1$). This indicates that the observed spread in H_0 is larger than expected from the quoted distance errors alone. The natural explanation is that galaxies have peculiar velocities, that is, motions on top of the Hubble expansion, which are not explicitly included in the distance error budget.
To account for this, I treat all eight $H_{0,i}$ as measurements of a single underlying H_0 with an additional intrinsic dispersion term $\sigma_{\rm int,H0}$:

$$\sigma^2_{H_0,i,\rm tot} = \sigma^2_{H_0,i} + \sigma^2_{\rm int,H0}. \tag{37}$$

I then fit a constant H_0 using these enlarged errors and choose $\sigma_{\rm int,H0}$ such that $\chi^2_\nu \approx 1$.
The resulting combined estimate is

$$H_0 = 67.5 \pm 5.8\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{38}$$

$$\sigma_{\rm int,H0} \simeq 16.2\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{39}$$

$$\chi^2_\nu \simeq 1.000. \tag{40}$$

It is noted that with a sample size of only $N = 8$, the uncertainty on this intrinsic dispersion estimate is significant (of order 25%), though its inclusion is necessary to obtain a realistic error budget for H_0. This value is also consistent within one standard deviation with the Planck 2018 cosmic microwave background analysis result of $H_0 = 67.4 \pm 0.5\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, although the uncertainty here is roughly an order of magnitude larger.
The intrinsic dispersion corresponds to velocity variations of a few hundred $\mathrm{km\,s^{-1}}$ at distances of order 10 to 20 Mpc, which is a reasonable scale for peculiar velocities.

4.3 Hubble diagram

A Hubble diagram was constructed by plotting recession velocity v versus distance D for all eight galaxies, with horizontal error bars indicating σ_D. A straight line through the origin with slope $H_0 = 67.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is overplotted.
Once the intrinsic dispersion is included, all galaxies are consistent with a common value of H_0 within their total uncertainties.

5 Summary

This report presents a complete analysis chain from Milky Way Cepheid parallaxes to a combined estimate of the Hubble constant. The key quantitative results are:

- **PL calibration (Step 1):**

$$M = -2.401\,(\log_{10} P - 0.845) - 3.636,$$

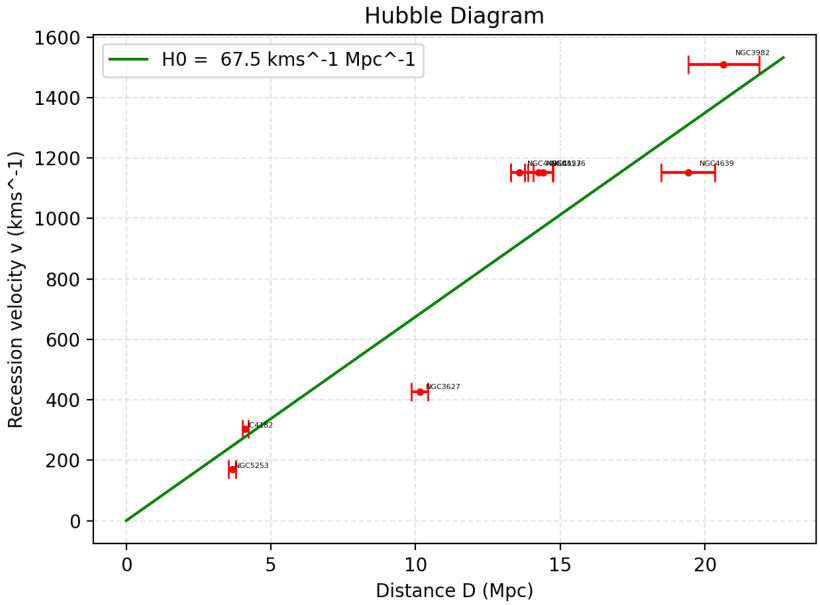


Figure 2: Hubble diagram for the eight galaxy sample. Each point shows the recession velocity v versus distance D, with horizontal error bars from the distance uncertainties. The green line shows the best fit relation $v = H_0 D$ with $H_0 = 67.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. *The scatter about the line is consistent with the combination of measurement errors and the inf*

with uncertainties of ± 0.232 on the slope and ± 0.049 mag on the intercept, reduced chi squared $\chi^2_\nu \simeq 0.535$, and negligible correlation between slope and intercept after shifting by $x_0 = 0.845$.

- **NGC4527 distance (Step 2):** after allowing for an intrinsic dispersion of about 0.30 mag in the Cepheid distance moduli,

$$\mu_{\mathrm{NGC4527}} = 30.768 \pm 0.072\,\mathrm{mag}, \qquad D_{\mathrm{NGC4527}} = 14.24 \pm 0.48\ \mathrm{Mpc},$$

with $\chi^2_\nu \simeq 0.960$.

- **Hubble constant (Step 3):** combining NGC4527 with seven additional galaxies, and including an intrinsic dispersion in H_0 to account for peculiar velocities,

$$H_0 = 67.5 \pm 5.8\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad \sigma_{\mathrm{int,H0}} \simeq 16.2\ \mathrm{km\,s^{-1}\,Mpc^{-1}},$$

with $\chi^2_\nu \simeq 1.000$. This value agrees within one standard deviation with the Planck 2018 result $H_0 = 67.4 \pm 0.5\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, though with a much larger uncertainty because of the small sample size and strong impact of peculiar velocities.

The main non standard analysis choices are:

1. introducing intrinsic dispersion terms when the initial reduced chi squared values were too large to be explained by the quoted errors alone; and

2. decorrelating the PL slope and intercept by shifting the $\log_{10} P$ values to a reference x_0 near their weighted mean.

Both of these are standard techniques in astronomical data analysis and are explicitly motivated by the behaviour of the χ^2 statistics for these datasets. Apart from these points, the analysis relies on straightforward weighted least squares fitting and Gaussian error propagation.